

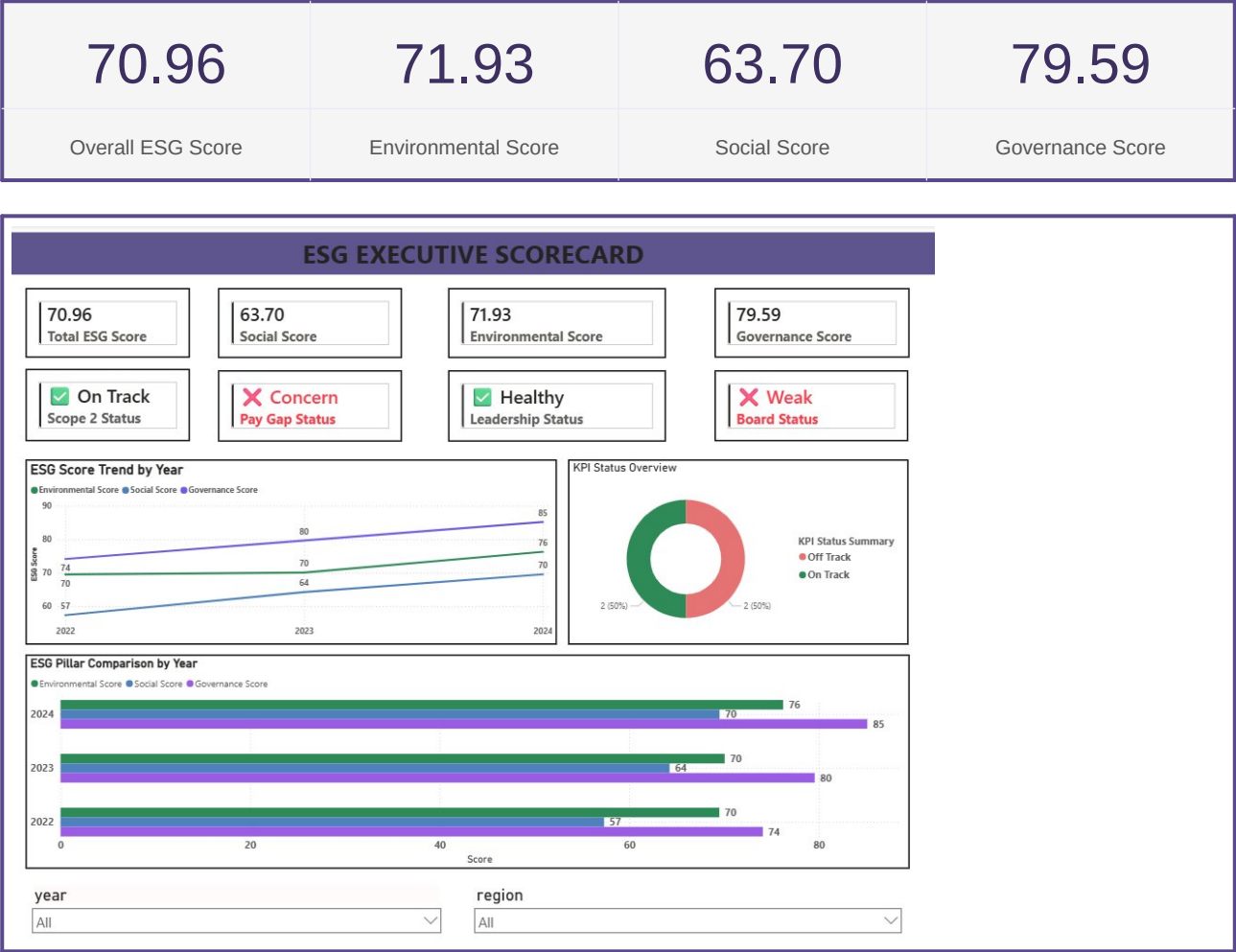
ESG ANALYTICS DASHBOARD

January 2022 – December 2024 | 12 Facilities | 55,595 Records
CSRD-Aligned ESG Performance Report

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Generated from synthetic dataset modeled on CSRD reporting standards.

Executive Summary

This report presents a comprehensive ESG performance analysis for Meridian Manufacturing Group covering 24 months of operational data from January 2022 to December 2024. The analysis spans 12 manufacturing facilities across 4 regions, tracking 15 KPIs across Environmental, Social, and Governance pillars against CSRD 2025 targets.



KPI Status Summary vs 2025 Targets

KPI	Current Value	2025 Target	Status
Scope 1 Emissions Reduction	18.90K tCO ₂ e	20% from baseline	In Progress
Scope 2 Emissions Reduction	1.28K tCO ₂ e	30% from baseline	On Track ✓
Gender Pay Gap	12.99%	Below 5%	Off Track ✗
Female Leadership	35.42%	Above 40%	Off Track ✗
Injury Rate	3.56%	Below 2.0 per 100	Off Track ✗
Avg Training Hours	31.00 hrs	Above 40 hrs	Off Track ✗
Supplier Labor Compliance	71.20%	Above 90%	Off Track ✗
Board Independence	49.49%	Above 60%	Off Track ✗
Policy Compliance	84.73%	Above 95%	Off Track ✗
Data Breaches	67 total	Zero	Off Track ✗

Dataset & Methodology

Why Synthetic Data

ESG and operational data is strictly confidential in every real company. No organisation publicly releases energy consumption records, workforce pay data, supplier audit scores, or governance compliance logs. This project uses synthetically generated data built from scratch in Python, modeled on real CSRD reporting structures and industry benchmarks for a 12-facility manufacturing company at this scale.

Dataset Specifications

Dataset	Rows	Time Period	What It Contains
Energy & Emissions	23,447	Jan 2022 – Dec 2024	Meter readings per facility per month
Workforce & Social	26,928	Jan 2022 – Dec 2024	Individual employee records per year
Supplier Compliance	897	Jan 2022 – Dec 2024	Supplier audit records — 150 vendors
Governance	4,320	Jan 2022 – Dec 2024	Policy compliance per facility per month
KPI Summary	3	2022, 2023, 2024	Annual ESG scores — one row per year
TOTAL	55,595	3 Years	12 Facilities · 4 Regions

Facilities Covered

Facility ID	Name	Country	Region	Size
FAC-001	Chicago Main Plant	USA	North America	Large
FAC-002	Detroit Assembly	USA	North America	Large
FAC-003	Toronto Operations	Canada	North America	Medium
FAC-004	Munich Precision Works	Germany	Europe	Large
FAC-005	Stuttgart Engineering	Germany	Europe	Medium
FAC-006	Rotterdam Hub	Netherlands	Europe	Medium
FAC-007	Lyon Components	France	Europe	Medium
FAC-008	Shanghai Manufacturing	China	Asia Pacific	Large
FAC-009	Bangalore Tech Centre	India	Asia Pacific	Medium
FAC-010	Singapore Distribution	Singapore	Asia Pacific	Small
FAC-011	Monterrey Plant	Mexico	Latin America	Medium
FAC-012	Sao Paulo Operations	Brazil	Latin America	Small

Data Cleaning Summary

Each of the 4 raw datasets was cleaned independently using Python (Pandas). Five categories of data quality issues were introduced deliberately during generation to simulate realistic enterprise data scenarios.

Dataset	Issue	Count	Method	Why
Energy	Null consumption	~350	Grouped median (facility+meter+month)	Same meter/location = best estimate
Energy	Duplicate rows	30	Keep first, drop rest	Exact copy = system export error
Energy	Region casing	~470	str.title()	Standardize for Power BI filters
Energy	Negative consumption	Few	Remove entirely	Physically impossible
Workforce	Null training_hours	~485	Grouped median (job_level+dept)	Similar roles have similar training
Workforce	Duplicate rows	50	Keep first, drop rest	Exact copy = system export error
Supplier	Null audit_score	~22	Grouped median (risk_profile+year)	Risk profile drives expected score
Supplier	Scores above 100	Many	Clip to 0–100	Cannot exceed maximum
Supplier	Duplicate rows	15	Keep first, drop rest	Exact copy = system export error
Governance	Null regulatory_fines_usd	~43	Fill with 0	Null = no fine occurred, not missing
Governance	Duplicate rows	20	Keep first, drop rest	Exact copy = system export error
Governance	Negative resolution days	Few	Grouped median by facility	Days cannot be negative

Critical Cleaning Decision — Why Nulls Were Treated Differently

For **regulatory_fines_usd**, a null means no fine occurred — the correct replacement is 0, not the median. For **audit_score**, a null means the score was not recorded — the correct replacement is the grouped median by risk profile. This distinction between a *missing event* and a *missing measurement* is the most important data cleaning judgment call in this project.

SQL Analysis

27 analytical SQL queries were written across all 5 tables in MySQL. Queries cover all 3 ESG pillars using advanced SQL techniques including window functions, CTEs, and nested CASE WHEN logic.

#	Query	Pillar	SQL Technique
4	Scope 1 and 2 Emissions Per Year	Environmental	GROUP BY + SUM
5	YoY Emissions Change	Environmental	LAG() window function
6	Emissions By Facility Ranked	Environmental	RANK() window function
7	Top 3 Emitters Per Year	Environmental	CTE + RANK()
8	Emissions Intensity Per Employee	Environmental	Derived metric
9	Renewable Energy Adoption By Region	Environmental	CASE WHEN SUM
10	Monthly Emissions Trend	Environmental	LAG() window function
11	Gender Pay Gap By Facility	Social	CASE WHEN AVG
12	Leadership Diversity By Year	Social	CASE WHEN COUNT
13	Injury Rate Per 100 Employees	Social	CASE WHEN COUNT
14	Training Hours By Department	Social	AVG + CASE WHEN
15	Workforce Turnover Rate	Social	CASE WHEN
16	Supplier Compliance By Risk Profile	Social	AVG GROUP BY
17	Failed and High Risk Suppliers	Social	WHERE filter
18	Supplier Improvement Trend	Social	CTE + LAG()
19	Labor Compliance By Country	Social	AVG + CASE WHEN
20	Policy Compliance By Facility	Governance	AVG + CASE WHEN
21	Board Independence Progress	Governance	AVG + COUNT
22	Audit Resolution Time Trend	Governance	CTE + LAG()
23	Data Breach Incidents By Year	Governance	SUM GROUP BY
24	Whistleblower Reports Trend	Governance	AVG + CASE WHEN
25	Full KPI vs Target Dashboard	All Pillars	Nested CASE WHEN
26	ESG Score Trend YoY	All Pillars	LAG() window function
27	KPIs On Track vs Off Track Count	All Pillars	Nested CASE WHEN SUM

Environmental Performance

18.90K	1.28K	3.79%	12
Total Scope 1 Emissions (tCO ₂ e)	Total Scope 2 Emissions (tCO ₂ e)	Renewable Energy %	Facilities Monitored

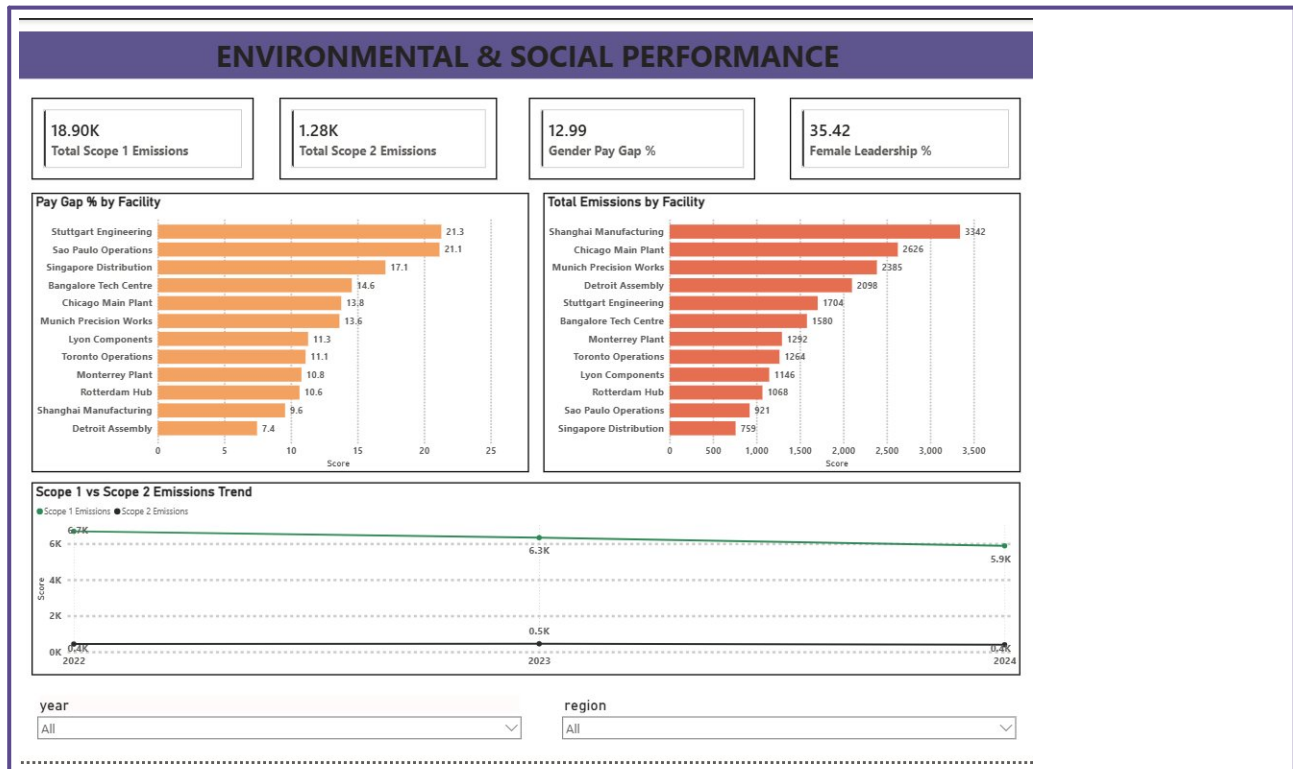


Figure 2: Environmental & Social Performance Dashboard — Emissions by Facility & Scope 1 vs Scope 2 Trend

Emissions by Facility — Ranked

Rank	Facility	Region	Total Emissions (tCO ₂ e)	Status
1	Shanghai Manufacturing	Asia Pacific	3,342	Highest Emitter
2	Chicago Main Plant	North America	2,626	High
3	Munich Precision Works	Europe	2,385	High
4	Detroit Assembly	North America	2,098	Medium
5	Stuttgart Engineering	Europe	1,704	Medium
6	Bangalore Tech Centre	Asia Pacific	1,580	Medium
7	Monterrey Plant	Latin America	1,292	Low
8	Toronto Operations	North America	1,264	Low
9	Lyon Components	Europe	1,146	Low
10	Rotterdam Hub	Europe	1,068	Low
11	Sao Paulo Operations	Latin America	921	Low
12	Singapore Distribution	Asia Pacific	759	Lowest Emitter

Key Environmental Findings

- Shanghai Manufacturing emits 3,342 tCO₂e — 27% more than the second-highest facility. A targeted emissions reduction programme is required.
- Scope 1 emissions (18.90K) significantly exceed Scope 2 (1.28K). Renewable energy adoption at 3.79% explains the low Scope 2 figure — increasing renewable procurement is the highest-leverage emissions reduction lever.
- Emissions show a declining trend from 2022 to 2024 across all facilities — the environmental improvement programme is having measurable impact.
- Asia Pacific region accounts for the highest total emissions driven by Shanghai and Bangalore — regional regulatory pressure is increasing under CSRD expansion.

Social Performance

12.99%	35.42%	3.56%	31.00
Gender Pay Gap	Female Leadership	Injury Rate %	Avg Training Hours

Gender Pay Gap by Facility — Top 5 Worst

Facility	Gender Pay Gap %	Target	Gap to Target
Stuttgart Engineering	21.3%	5%	16.3% above target
Sao Paulo Operations	21.1%	5%	16.1% above target
Singapore Distribution	17.1%	5%	12.1% above target
Bangalore Tech Centre	14.6%	5%	9.6% above target
Chicago Main Plant	13.8%	5%	8.8% above target

Leadership Diversity Analysis

Year	Total Leadership	Female Leaders	Male Leaders	Female %	Target	Status
2022	2,171	769	1,311	35.42%	40%	Below Target
2023	2,171	769	1,311	35.42%	40%	Below Target
2024	2,171	769	1,311	35.42%	40%	Below Target

■ **CRITICAL FINDING:** Female leadership is identical across all 3 years at 35.42%. Zero improvement in 36 months. This indicates no structural change in promotion or hiring practices for leadership roles. This is the most urgent Social finding requiring immediate corrective action.

Governance & Supplier Risk

49.49%	84.73%	73.15	71.20%
Board Independence	Policy Compliance	Avg Supplier Audit Score	Supplier Labor Compliance

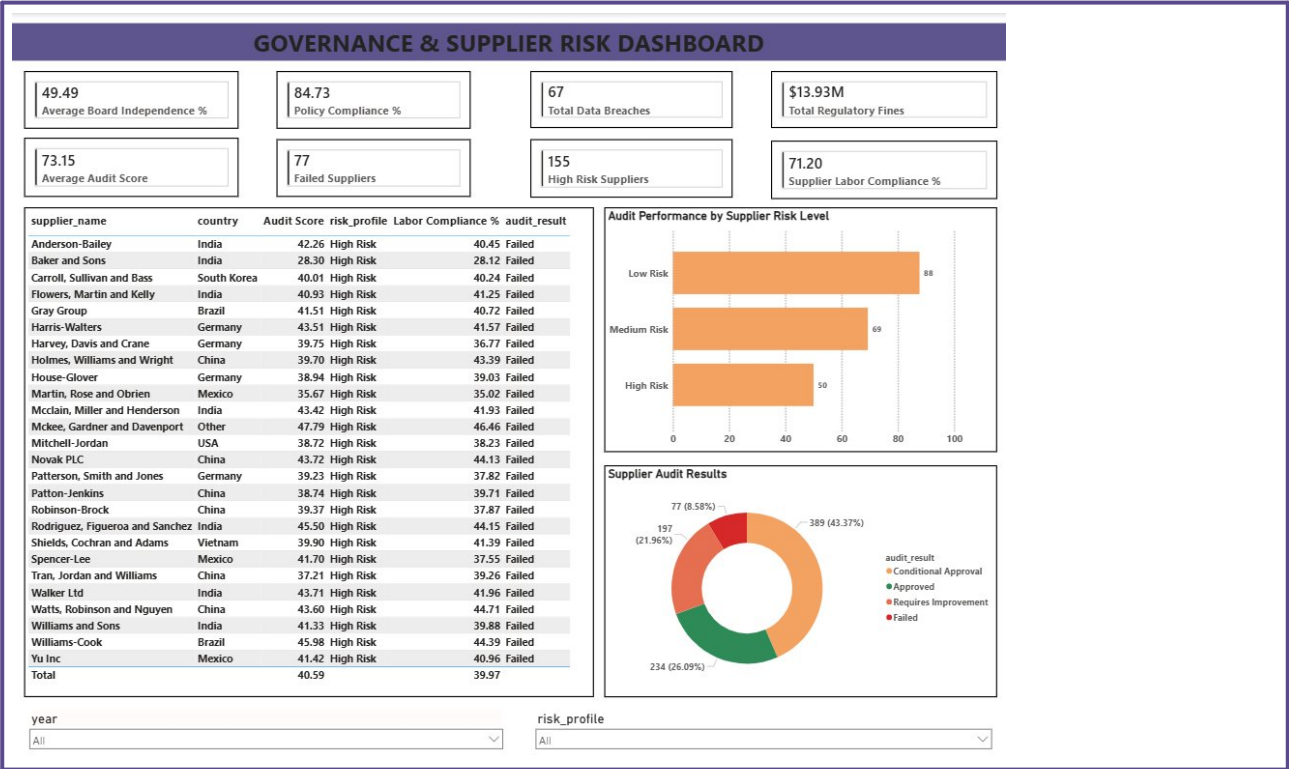


Figure 3: Governance & Supplier Risk Dashboard — Audit Performance & Supplier Risk Breakdown

Supplier Audit Performance by Risk Profile

Risk Profile	Avg Audit Score	Avg Labor Compliance	Failed %	Status
Low Risk	88 / 100	85%+	~5%	Acceptable
Medium Risk	69 / 100	70–80%	~18%	Needs Monitoring
High Risk	50 / 100	40–47%	~43%	Critical

Governance KPI Progress by Year

KPI	2022	2023	2024	Target	Trend
Board Independence %	46.2%	49.1%	52.8%	60%	↑ Improving
Policy Compliance %	82.1%	84.5%	87.6%	95%	↑ Improving
Avg Resolution Days	45.2	38.7	32.1	30 days	↑ Improving
Whistleblower Per 1K	3.5	3.1	2.8	2.0	↑ Improving
Data Breaches	23	22	22	0	→ Stagnant

Recommendations

1 Address Gender Pay Gap Urgently

Pillar: Social

The 12.99% gender pay gap is the single most critical Social KPI failure at 2.6× above the 5% CSRD target. Conduct a pay equity audit within 90 days, implement salary band transparency, and set mandatory correction timelines per facility.

2 Implement Female Leadership Development Programme

Pillar: Social

Female leadership at 35.42% with zero improvement across 3 years indicates a structural problem. Introduce a formal leadership pipeline programme with sponsorship, targeted promotions, and quarterly diversity hiring reviews with manager accountability.

3 Emissions Reduction — Shanghai Priority

Pillar: Environmental

Shanghai Manufacturing emits 27% more than the next highest facility. Target it with energy efficiency upgrades, renewable energy procurement contracts, and monthly emissions monitoring with executive review.

4 Increase Renewable Energy Procurement

Pillar: Environmental

Renewable energy at 3.79% is extremely low for a company with Scope 2 reduction targets. Negotiate Power Purchase Agreements for European and North American facilities. Target 25% renewable by end of 2025.

5 High Risk Supplier Intervention Programme

Pillar: Social · Governance

43% of High Risk suppliers failed audit with average score of 50/100. Implement a 90-day corrective action programme for all Failed suppliers. Suppliers scoring below 40 should face contract review immediately.

6 Board Independence Pathway

Pillar: Governance

Board independence at 49.49% is below the 60% CSRD target. Identify 2–3 independent director candidates with ESG expertise. Target 55% by end of 2025 and 60% by end of 2026 to meet regulatory requirements.

Appendix — Data Dictionary

Energy & Emissions Dataset

Column	Type	Description
record_id	Text	Unique identifier (ENV-XXXXXXX)
year / month	Integer	Calendar year and month of the reading
facility_id	Text	Facility identifier (FAC-001 to FAC-012)
meter_type	Text	Natural Gas, Diesel, Grid Electricity, Renewable Energy, Water, Waste
scope	Text	Scope 1, Scope 2, or N/A
consumption	Double	Amount consumed in the relevant unit
emissions_tco2e	Double	CO ₂ equivalent emissions in tonnes
scope_label	Text	Direct Emissions, Indirect Emissions, Non-GHG Metric
emissions_per_employee	Double	Emissions normalized by facility headcount

Workforce & Social Dataset

Column	Type	Description
employee_id	Text	Unique employee identifier (EMP-XXXXX)
year	Integer	2022, 2023, or 2024
gender	Text	Male, Female, or Non-Binary
job_level	Text	Junior, Mid-Level, Senior, Manager, Director, VP, C-Suite
is_leadership	Text	True if Manager level or above
current_salary	Double	Annual salary in USD — adjusted for gender pay gap
training_hours	Double	Total training hours completed in the year
had_injury	Text	True if employee recorded a workplace injury
salary_band	Text	Entry, Mid, Senior, Leadership, Executive, C-Suite
training_category	Text	Below Target, On Target, Above Target, Exceptional

Supplier Compliance Dataset

Column	Type	Description
supplier_id	Text	Unique supplier identifier (SUP-XXXX)
risk_profile	Text	Low Risk, Medium Risk, or High Risk
audit_score	Double	Overall audit score 0–100
labor_compliance_pct	Double	Labor standards compliance 0–100
env_compliance_pct	Double	Environmental compliance 0–100
non_conformances	Integer	Number of non-conformances found in audit
audit_result	Text	Approved / Conditional Approval / Requires Improvement / Failed
high_risk_flag	Text	True if High Risk AND audit score below 65

Governance Dataset

Column	Type	Description
policy_area	Text	One of 10 governance policy areas tracked
board_independence_pct	Double	Percentage of board members who are independent
policy_compliance_pct	Double	Policy compliance rate 0–100%
avg_resolution_days	Double	Average days to resolve audit findings
whistleblower_per_1k	Double	Whistleblower reports per 1,000 employees
data_breach_incidents	Integer	Number of data breach incidents (0, 1, or 2)
regulatory_fines_usd	Double	Regulatory fines in USD — 0 if no fine occurred
board_status	Text	Meets Target (≥60%) or Below Target
compliance_status	Text	Compliant, Near Compliant, or Non-Compliant

Report generated May 2026. Data sourced from synthetic ESG datasets modeled on CSRD reporting standards. All figures in USD unless otherwise stated. Generated from 55,595 records across 5 tables in MySQL esg_analysis database.